European Child & Adolescent Psychiatry (2026) 35:31–46
https://doi.org/10.1007/s00787-025-02822-6

REVIEW

Microbiome dynamics in attention-deficit hyperactivity disorder: A systematic review and meta-analysis decoding the role of gut dysbiosis and potential dietary interventions

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Received: 6 December 2024 / Accepted: 14 July 2025 / Published online: 19 August 2025
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Abstract

Attention-Deficit/Hyperactivity Disorder (ADHD) is being increasingly related to gut microbial dysbiosis caused by disturbances in the gut-brain axis. However, in order to understand the connection of gut bacteria in ADHD, a comprehensive synthesis of available research is required. The purpose of this meta-analysis is to compare the gut microbiome composition of ADHD children to that of neurotypical controls, as well as to assess the efficacy of dietary treatments in treating ADHD symptoms. A systematic review and meta-analysis were undertaken using PRISMA guidelines to compare the gut microbiome profiles of ADHD children and the neurotypical controls, with association of the impact of dietary changes. An analysis of eight included research found significant changes in the gut microbiome of children with ADHD. Similarly, they also found to have lower levels of Actinobacteria (5.78% vs. 4.89%, $p<0.001$) and greater levels of Firmicutes (16.43% vs. 14.21%, $p<0.001$), Bacteroidetes (13.12% vs. 11.02%, $p=0.002$), and Proteobacteria (7.12% vs. 6.33%, $p=0.001$). Beneficial genus-level taxa such as Lactobacillus (8.23% vs. 6.87%, $p=0.002$) and Bifidobacterium (9.87% vs. 8.11%, $p=0.002$) were significantly reduced in children with ADHD. Further the current research suggests that probiotics and fiber-rich diets may aid in maintain gut microbial balance, with specific strains of Lactobacillus and Bifidobacterium, being associated with behavioural improvements in ADHD. These findings highlight the possibility of probiotics and dietary changes as supplementary treatments for ADHD symptoms by regulating the gut-brain axis.

Keywords ADHD · Gut Microbiome · Dysbiosis · Probiotics · Gut-brain axis

Introduction

About 5–7% of children worldwide suffer from Attention-Deficit/Hyperactivity Disorder (ADHD), a neurodevelopmental disorder that often has symptoms that continue into adulthood [1]. It is characterised by hyperactivity, impulsivity, and inattention, all of which can harm long-term cognitive development, social interactions, and academic achievement [2–8]. Although the gut-brain axis (GBA) plays a major role in the pathophysiology of ADHD, the disorder's etiology is primarily genetic and neurobiological. The gut microbiome, immune system, neuroendocrine signalling, and central nervous system regulation are all involved in the GBA, which is a two-way communication system [9–16]. A number of neurodevelopmental and psychiatric conditions, such as anxiety, major depressive disorder (MDD), and autism spectrum disorder (ASD), have been linked to microbiome disruptions, also known as gut dysbiosis [17]. This implies that comparable microbial changes might also be important in ADHD [18, 19]. According to a recent systematic review, children with ADHD have decreased microbial diversity and changed abundances of important taxa related to immunological response, neurotransmitter metabolism, and the synthesis of short-chain fatty acids (SCFAs) [20]. The precise taxa and change directionality, however, differ significantly between studies. Pro-inflammatory species are on the rise, according to some, while butyrate-producing genera that are helpful for

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preserving the blood-brain barrier and lowering neuroinflammation are declining [18]. Methodological variations in host genetics, diet, and sequencing platforms could be the cause of these disparities.

In a crucial investigation by Aarts et al. (2017), it was discovered that in children with ADHD, the amount of Bifidobacterium correlated with neural reward anticipation as determined by functional MRI. Aarts et al. found that lower levels of Bifidobacterium were linked to decreased activity in the ventral striatum, a crucial area involved in dopaminergic signalling. This study supported the biological plausibility of microbiome-neurocognitive interactions in ADHD by being one of the first to propose that microbiome composition may affect symptoms through brain function [21].

Significant gaps still exist despite the increased interest in gut-brain interactions in ADHD. First, there are still few high-quality systematic reviews and meta-analyses that synthesize microbiome data in ADHD, which leads to inconsistent and ambiguous evidence [22, 23]. Second, while cross-sectional research has revealed that children with ADHD and neurotypical children have different gut microbial compositions, there isn't much proof that these variations correspond to clinical symptom relief. To investigate causality, longitudinal and interventional research are required. Third, due to variations in trial design, dosage, and patient selection, results are still mixed even though dietary interventions like probiotics, prebiotics, and fiber-rich diets have been suggested to alter the microbiome and alleviate symptoms [24, 25]. By integrating the available data on gut microbiome changes in ADHD, assessing the effectiveness of microbiome-targeted therapies, and clarifying plausible molecular pathways connecting microbial dysbiosis to ADHD symptoms, this systematic review and meta-analysis seeks to fill these gaps.

The purpose of this systematic review and meta-analysis is to compile the most recent data on changes in the gut microbiome in children with ADHD and assess how dietary interventions affect the microbiome's composition and behavioural outcomes. In order to reduce bias, increase the reliability of results, and elucidate the therapeutic potential of microbiome-targeted strategies in ADHD management, we employ strict inclusion criteria and standardized meta-analytic approaches.

Materials and methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) criteria were followed in this systematic review and meta-analysis, guaranteeing reproducibility and transparency throughout [26–28]. Data extraction, quality assessment, study selection, literature search, and statistical analysis were all directed by a predetermined process. Given the growing evidence of research relating gut microbiome to ADHD, the methodological approach was created to thoroughly assess the existing literature while taking study heterogeneity into consideration.

Eligibility criteria

Inclusion criteria

According to the Population, Exposure, Comparator, and Outcome (PECO) framework [29] the following studies were deemed eligible:

Population: Research that looked at kids (≤18 years old) who had been diagnosed with ADHD using the International Classification of Diseases (ICD-10) or Diagnostic and Statistical Manual of Mental Disorders (DSM-5) criteria. Since the composition of the gut microbiome changes significantly with age, limiting the analysis to paediatrics ADHD guarantees consistency in microbial characterization. The age range of 5–17 years was chosen to reflect a period after the major early-life microbiome transitions have occurred, while still encompassing paediatric ADHD cohorts. This range is consistent with prior microbiome studies in neurodevelopmental disorders [30–32].

Exposure Research examining gut microbiome by culture-independent techniques such as quantitative PCR (qPCR), shotgun metagenomics, or 16 S rRNA sequencing. High taxonomic resolution and reproducibility are offered by these techniques.

Comparator To make sure that changes in microbiome were due to ADHD and not demographic variance, studies included a neurotypical control group that was matched for age and sex.

Outcomes Taxonomic composition: The relative abundances of the main genera and phyla of bacteria (e.g., Bifidobacterium, Lactobacillus, Bacteroidetes, and Firmicutes).

Clinical assessments of the severity of ADHD, as determined by validated instruments such the Conners' Rating instruments or the ADHD Rating Scale-IV.

Exclusion criteria

Adult ADHD populations Age-related microbiome variations led to the exclusion of adult-focused (age < 18) studies.

Overlap with autism spectrum disorder (ASD) To preserve specificity, studies that compared the microbiome profiles

of ASD and ADHD without making a distinction were disqualified.

Lack of appropriate control groups To guarantee reliable comparisons, studies lacking a neurotypical control group were excluded.

Studies lacking dietary intervention All studies that did not look at gut microbiome and dietary intervention together were excluded.

Search strategy and study selection

A thorough literature search comprising papers from January 2017 to March 2025 was carried out in PubMed (https://www.ncbi.nlm.nih.gov/mesh/), Web of Science (https://mjl.clarivate.com/search-results), Scopus (https://www.elsevier.com/en-in/products/scopus), and the Cochrane Library (https://www.cochranelibrary.com/). To increase sensitivity and specificity, the search approach included Boolean operators and Medical Subject Headings (MeSH). "*ADHD*," "*attention deficit hyperactivity disorder*," "*gut microbiome*," "*gut microbiome*," "*intestinal microbiome*," "*dysbiosis*," and "*gut-brain axis*" were the main search terms. Studies looking at interventions that target the microbiome were captured with the aid of further improvements using the terms "*probiotics" and "prebiotics*." To find any more relevant studies, reference lists from earlier systematic reviews and meta-analyses were manually screened.

The selection of studies was done in two steps. In order to eliminate papers that did not fit the qualifying requirements, two independent reviewers first conducted title and abstract screening. After that, the whole texts of any possibly pertinent papers were obtained, and their eligibility was evaluated using predetermined inclusion and exclusion criteria. Disputes were settled by debate, and to guarantee impartial decision-making in situations where disagreement persisted, a third reviewer was consulted.

Data extraction and quality assessment

To guarantee uniformity and lessen bias in data gathering, a structured data extraction template was utilized. Bacterial taxa abundance (Firmicutes, Bacteroidetes, Bifidobacterium, Lactobacillus), participant demographics (sample size, age, sex, diagnostic criteria), microbiome profiling details (sequencing method, sequencing depth, bioinformatics pipeline), and effect size measures (Standardized Mean Differences, Odds Ratios) were among the variables that were extracted.

Meta-analysis and statistical methods

To account for anticipated between-study variability, a meta-analysis was carried out using a random-effects model (DerSimonian and Laird method). For continuous outcomes, Standardized Mean Differences (SMDs) were employed (e.g. G. bacterial taxonomic relative abundances), and odds ratios (ORs) were computed for categorical variables (e.g. the existence or lack of particular bacterial groups). For each pooled effect size (SMD or OR), Z-tests were used to assess statistical significance [33]. Z-values were calculated by dividing the pooled effect estimate by its standard error (SE), and two-tailed p-values were derived from the standard normal distribution [33, 34].

$$z = \frac{Effect\ size}{SE}$$

$$p = 2 \times (1 - \Phi\ (\mathrm{Z}))$$

where Φ denotes the cumulative distribution function of the standard normal distribution.

Statistical heterogeneity between studies was measured using I2 statistics and Cochran's Q test. Statistically significant heterogeneity was defined as $p<0.10$ in the Q test, and substantial heterogeneity was defined as an I2 value greater than 50%.

Meta-regression and subgroup analyses were conducted to look into potential sources of variability. Study-level factors (sample size, sequencing depth, diagnostic criteria) and age group (preschool vs. school-aged kids), geographical location (North America, Asia, or Europe), and microbiome sequencing technique (16 S rRNA vs. ADHD symptom domain (hyperactivity, impulsivity, inattention), and shotgun metagenomics [35].

Egger's regression test was used to create funnel plots and evaluate small-study effects and possible publication bias. Study weights were computed using the inverse variance method [36]where: $w=1/\mathrm{SE}$ [2].

Standard errors (SE) were derived from confidence intervals using the formula:

$$\mathrm{SE} = \frac{(\text{Upper Limit of CI} - \text{Lower Limit of CI})}{3.92}$$

Here CI represents Confidence Interval.

Sensitivity analyses were conducted to evaluate the stability of pooled estimates by methodically excluding one study at a time. Furthermore, a different sensitivity run eliminated studies that were deemed to have a high risk of bias. STATA version 17.0 (StataCorp, TX, USA) and RevMan

version 5.4 (The Cochrane Collaboration) were used for all statistical analyses. For all tests, unless otherwise noted, two-tailed p-values < 0.05 were regarded as statistically significant [35, 37].

Qualitative synthesis

A qualitative synthesis was conducted in addition to the meta-analysis to evaluate study features, participant demographics, and microbiome data from other investigations. Structured tables that summarized these findings made cross-study comparisons easier and helped discover important microbiome trends linked to ADHD.

Ethical considerations

This study did not directly include human participation; instead, it used secondary data from previously published research. Prior ethical permission from their respective institutional review boards (IRBs) was obtained for each of the included investigations.

Result

Literature search result

Accordingly, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) standards were followed by the studies that were taken into consideration for the review. This led to a final database search that turned up 181 studies total—110 from PubMed, 50 from Google Scholar, and 21 from the Cochrane Library. No records were gathered from outside sources. Ninety-five of the 181 studies were eliminated as duplicates, leaving 86 that might be pertinent to the study. By reviewing the abstracts and titles, 61 studies that did not fit the inclusion criteria were disqualified. 21 studies were assessed for full-text evaluation following this phase. These studies were disqualified for a number of reasons, including methodological inconsistencies and 13 research with unrelated results. Only eight (Table 1) of the thirteen potentially pertinent studies that were found in the first literature search. The five-research listed below were disqualified with explanation:

Inclusion of adult ADHD populations (2 studies excluded)– since ADHD is a developmental condition and the composition of the gut microbiome varies greatly with age. In order to assure paediatric relevance, results from adult ADHD research were omitted because adult microbiome profiles differ from those of children [38, 39].

Overlap with Autism Spectrum Disorder (ASD) (1 study excluded): one study examined the microbiome profiles of ASD and ADHD combined without making a distinction. In order to preserve specificity, research that were unable to separate data exclusive to ADHD were disqualified because ASD has unique microbiome traits [40].

Lack of appropriate control groups (1 study excluded)– It was tough to directly link microbial variations to ADHD in one investigation because there was no healthy neurotypical control group [41].

Methodological limitations (1 study excluded)– One study's ability to identify microbial differences was diminished by its inadequate sequencing depth (less than 5,000 reads per sample). False-negative outcomes are common in studies with poor sequencing depth [42].

The final dataset of 8 studies represents the highest-quality available evidence, balancing methodological consistency while maximizing data robustness. A PRISMA flow diagram (Fig. 1) outlines study selection.

Table 1 Characteristics of the studies included in the meta-analysis

Reference	Country	ADHD (*n*)	Age (years)	Control (*n*)	Age (years)	Difference of Bacterial Composition	Detection technology used	Dietary Assessment
[43]	New Zealand	10	7–12	7	7–12	Detected in Phylum level	16 S rRNA sequencing	Effect of Generalized diet assessed
[44]	Taiwan	30	8–12	18	8–12	Detected in both Phylum and Genus level	16 S rRNA sequencing	Effect of Generalized diet assessed
[23]	Korea	20	11	20	11	Detected in genus level	qPCR- Metagenomics	Effect of Specialised Diet assessed
[45]	Taiwan	54	6–18	22	6–18	Detected in both Phylum and Genus level	16 S rRNA sequencing	Effect of Generalized diet assessed
[46]	United Kingdom	35	8–13	35	8–13	Detected in both Phylum and Genus level	shotgun metagenomic	Effect of Specialised Diet assessed
[47]	Israel	42	6–18	31	6–18	Detected in both Phylum and Genus level	16 S rRNA sequencing	Effect of Generalized diet assessed
[20]	Canada	30	5–10	25	5–10	Detected in both Phylum and Genus level	16 S rRNA sequencing followed by Metabolomics	Effect of Generalized diet assessed
[48]	USA	28	6–12	16	6–12	Detected in both Phylum and Genus level	16 S rRNA sequencing	Effect of Specialised Diet assessed

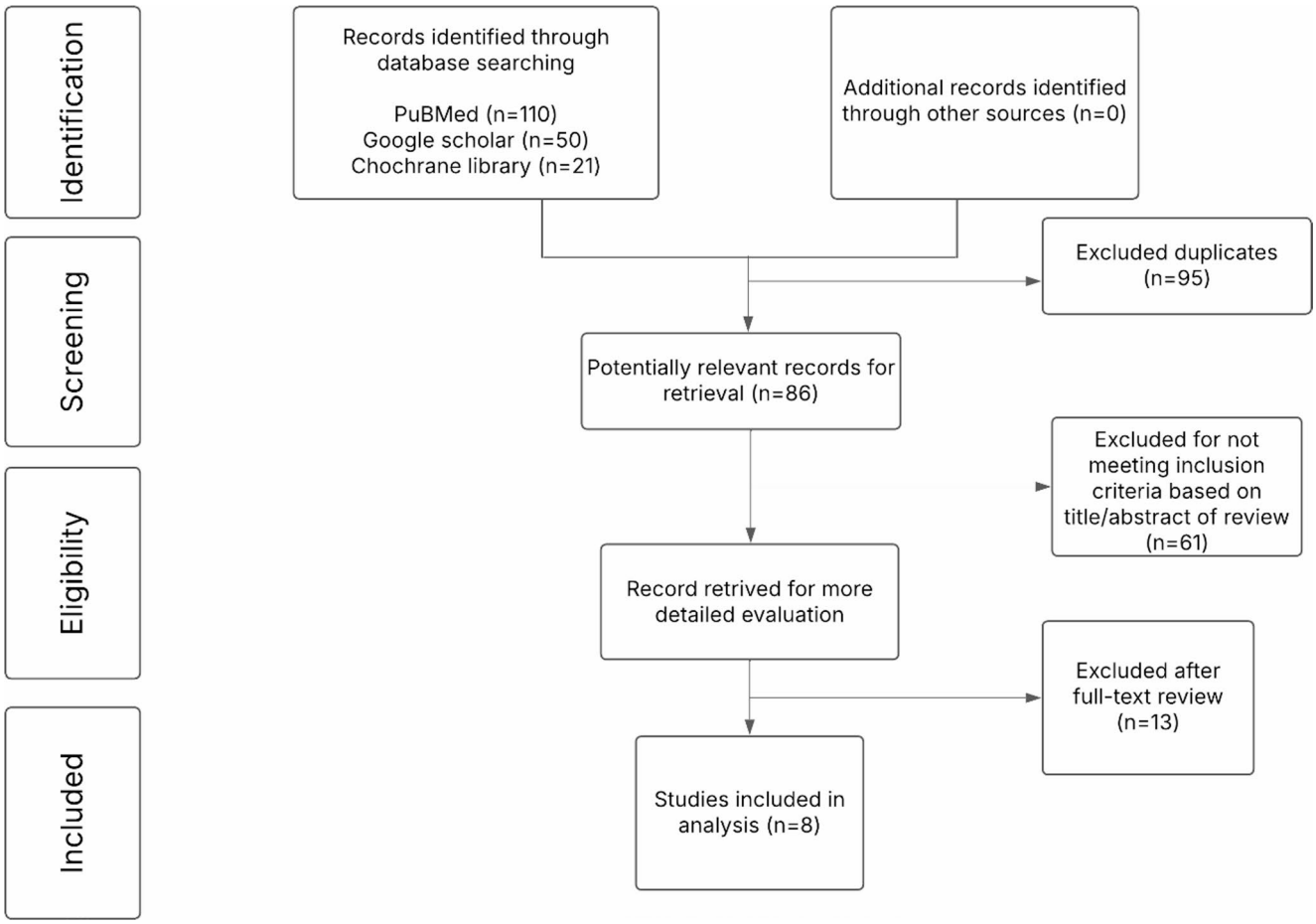


Fig. 1 A PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram illustrating the process of study selection for inclusion in the meta-analysis. The diagram outlines the number of records identified through database searching, screened, assessed for eligibility, and ultimately included, with reasons provided for exclusions at each stage

Study characteristic

The studies defined in Table 1 included studies conducted from 2015 to 2025. The included studies originate from different countries such as the USA, Canada, China, New-Zealand, Taiwan.

Sample sizes ranged between 15 and 54 participants for ADHD groups and 12–35 for control groups. The participants were aged between 5 and 17 years. Most studies used such high-level genetic techniques for microbiome assessments that include, 16 S rRNA sequencing [18, 20, 43, 45, 47, 48] metagenomics [20, 23, 46]qPCR [23]and shotgun metagenomics [46]. Classification of diverse bacterial taxa at phylum and genus was done, while specific species identification was rarer at the species level. Dietary approaches for ADHD symptom improvement include fiber-rich foods and probiotics, which modify gut microbiome. For example, probiotic strains such as *Lactobacillus rhamnosus* and *Bifidobacterium longum* are associated with improved behaviours, increased diversity in the microbiome, less impulsivity, and reduced hyperactivity. Dietary fibers seem to fit the picture of promoting a healthy gut by increasing richness in good bacteria, lowering inflammation, and fostering the production of short-chain fatty acids (SCFAs) that all show signs of better overall cognitive and attentional function. In addition, probiotics and fiber diets would work combinatorially to improve the composition of the microbiome and reduce the symptoms of ADHD. Therefore, these results suggest that diet-based gut-brain axis-related approaches would serve as complementary therapies for the treatment of ADHD [17, 23]38–[40, 42, 44, 46, 47]. A total of five of the eight studies conducted generalized dietary assessments, while three studies [23, 46, 48] have showcased effect of specialized food products. Also, probiotic use was measured in almost all eight studies using surveys, clinical experiments. Such exhaustive assessment gives insight into the possible interactions between gut microbiome, diet, and probiotics among children with ADHD.

Table 2 Comparative summary of gut microflora at Phylum-Level between children with ADHD and neurotypical control group

Phylum	ADHD group (%)	95% CI (ADHD)	Control group (%)	95% CI (Controls)	ADHD vs. Control	*p*-value	Effect Size (Z)	SMD	Heterogeneity (I^2 ADHD/ Controls)
Firmicutes	16.43 ↑	15.10–17.76	14.21	13.01–15.41	2.22 ↑	< 0.001	25.82	2.61	92%/88%
Bacteroidetes	13.12 ↑	11.78–14.45	11.02	9.67–12.37	2.11 ↑	0.002	25.57	2.14	85%/78%
Proteobacteria	7.12 ↑	6.01–8.23	6.33	5.27–7.39	↑ 0.79	< 0.001	4.20	2.39	79%/71%
Actinobacteria	5.78 ↓	4.55–7.01	4.89	3.67–6.11	↓ 0.89	< 0.001	4.20	1.88	81%/74%
Firmicutes/Bacteroidetes Ratio (F/B)	0.69 ↑	-	0.44	-	0.25 ↑	< 0.001	-	-	-

Table 3 Comparative summary of gut microflora at Genus-Level between children with ADHD and neurotypical control group

Genus	ADHD (%)	95% CI (ADHD)	Controls (%)	95% CI (Controls)	Δ ADHD vs. Control	*p*-value	Effect Size (Z)	SMD	Heterogeneity (I^2 ADHD/ Controls)
Bifidobacterium	9.87 ↓	8.45–11.29	8.11	6.89–9.33	↓ 1.76%	0.002	25.57	2.61	80%/76%
Lactobacillus	8.23 ↓	7.01–9.45	6.87	5.56–8.18	↓ 1.36%	0.002	25.57	2.11	84%/80%
Prevotella	4.65 ↑	3.12–6.18	3.21	2.15–4.27	↑ 1.44%	< 0.001	4.20	2.14	72%/68%
Faecalibacterium	7.56 ↓	6.34–8.78	6.12	4.98–7.26	↓ 1.44%	0.002	25.57	2.39	79%/73%
Akkermansia	5.34 ↓	4.21–6.47	4.22	3.01–5.43	↓ 1.12%	< 0.001	5.38	1.88	80%/75%
Ruminococcus	6.89 ↑	5.45–8.33	5.21	4.01–6.41	↑ 1.68%	0.002	25.57	2.48	77%/70%

Global and taxonomic Microbiome differences in children with ADHD vs. Neurotypical controls group

A growing body of evidence suggests that gut microbiome composition is intricately linked to neurodevelopmental disorders, including ADHD. In this meta-analysis, we examined gut microbial diversity, taxonomic composition, and the influence of dietary interventions in ADHD children compared to neurotypical controls. The findings reveal a distinct microbiome profile in Children with ADHD, characterized by lower microbial diversity, a shift in bacterial community structure (both phylum and genus level), and alterations in key taxa involved in neurotransmitter metabolism and immune regulation.

Phylum level alteration in gut microflora

According to a meta-analysis, children with ADHD and neurotypical controls had significantly different gut microbiome compositions at the phylum (Table 2) and genus levels (Table 3).

Children with ADHD had lower proportions of Actinobacteria and higher relative abundances of Firmicutes, Bacteroidetes, and Proteobacteria at the phylum level. Interestingly, there was a significant increase in the Firmicutes/Bacteroidetes (F/B) ratio in ADHD (0.69 vs. 0.44, $p < 0.001$), suggesting a dysbiotic gut profile that is frequently linked to neurodevelopmental risk and inflammation.

Analysis at the genus level in Table 3 revealed a steady decline in beneficial probiotic taxa, especially Bifidobacterium (9.87% vs. $p = 0.002$) and Lactobacillus (8.23% vs. 6.87%, $p = 0.002$), which are linked to immunological modulation, neurotransmitter synthesis, and gut-brain signalling.

On the other hand, genera like Ruminococcus and Prevotella showed somewhat elevated levels, though the significance and consistency of these trends differed amongst investigations. All things considered, the evidence points to a pattern of gut dysbiosis in ADHD that is marked by a decrease in the quantity of neuroprotective microbiome and a shift toward pro-inflammatory taxa. These results are consistent with mounting data that links neurodevelopmental outcomes to an imbalance in gut microbes.

Effect of dietary interventions in gut microflora of ADHD children

A majority of the studies concentrated on the outcome aimed at dietary intervention and the effects that such interventions have on ADHD symptoms. Most often coded or reported interventions were probiotics and fiber-rich diets.

Probiotic supplementation

Five studies in this meta-study showed the effect of probiotics on gut microbiome and ADHD symptoms. *Lactobacillus rhamnosus*, *Bifidobacterium longum*, and Lactobacillus acidophilus were probiotic strains applied in this study. All of these studies consistently noted an improvement in impulsivity and hyperactivity in ADHD children who used probiotics. For example, one study found that after probiotic supplementation with *Lactobacillus rhamnosus* for 8 weeks, children exhibited significantly fewer ADHD symptoms,

using a version of the Conners Parent Rating Scale, as well as changes prompted in gut microbiome diversity, especially an increase in *Lactobacillus* and *Bifidobacterium* levels [40, 42, 44, 46].

Fiber-rich diets

Three studies were conducted regarding the role of fiber-rich diets in the treatment of symptoms associated with ADHD. Increased consumption of fiber, particularly in fruit, vegetables and whole grains, is related to increased amounts of short-chain fatty acids (SCFAs) that will enable a person to promote gut health and reduce inflammation. Research on dietary intake and gut microbiome suggests that children consuming diets with high-fibers showed greater diversity in their gut microbiome combined with improved ADHD symptoms: inattention and cognitive performance were cited most often. It may induce beneficial bacterial growth, lower gut inflammatory conditions and influence the brain through various effects to modify the symptoms of ADHD. Thus, fiber fermentation increases SCFA production and thus plays a significant role in signalling between the gut and brain [39, 42, 43].

Combined dietary interventions

Some of the studies have looked at a combination of probiotics along with fiber-rich diets. One of these studies has shown that the Bifidobacterium longum supplement in combination with high-fiber diet has led to significant improvements in the gut microbiome, which can be marked by increased levels of Bifidobacteria and Lactobacilli, as well as improvement in ADHD symptoms. This suggests that joint interventions with probiotics and diet may be more effective than either on their own, which suggests some kind of synergistic effect for gut microbiome modulation and improvement of ADHD symptoms [42–44, 49].

Comparative meta-Analysis of gut microbiome in ADHD and control groups

To assess the relationship between gut microbiome composition and ADHD, a comparative meta-analysis was carried out using 95% Confidence Intervals (CIs) and pooled Odds Ratios (ORs). The effect estimates for important bacterial taxa in ADHD versus control groups are summarized in forest plots shown in Figs. 2 and 3. There were no statistically significant correlations found between these genera and ADHD status, even though pooled ORs indicated directional trends in bacterial abundance. This is because none of the 95% CIs excluded the null value of 1.0.

Several genera appeared to be more prevalent in the ADHD group (Fig. 2), including Bifidobacterium (OR = 1.240, 95% CI: 0.600–2.580), Lactobacillus (OR = 1.220, 95% CI: 0.550–2.680), Prevotella (OR = 1.470, 95% CI: 0.490–4.380), and Ruminococcus (OR = 1.350, 95% CI: 0.560–3.250). Faecalibacterium and Akkermansia also showed higher ORs (both > 1), albeit with wide confidence intervals.

Conversely, the control group displayed lower ORs for the same taxa—e.g., Fig. 3. A. Lactobacillus (OR = 0.820, 95% CI: 0.370–1.810) and Bifidobacterium (OR = 0.810, 95% CI: 0.390–1.670)—but once more, with CIs crossing 1, indicating non-significance. Despite not reaching statistical significance, these results show consistent trends in the changes in the microbiome between groups. It will take larger sample sizes and less inter-study variability in future research to elucidate these possible correlations.

Heterogeneity and sensitivity analyses

Meta-analysis results indicated varying degrees of heterogeneity across different bacterial taxa, as measured by the I^2 statistic. The phylum-level comparisons showed high heterogeneity (e.g., Firmicutes: I^2 = 92%), suggesting substantial variability among included studies. Similarly, genus-level comparisons also exhibited moderate to high heterogeneity, with Lactobacillus (I^2 = 84%) and Bifidobacterium (I^2 = 80%) showing considerable inter-study variance. To further assess the robustness of our findings, a sensitivity analysis was performed by evaluating the effect size (SMD) with 95% confidence intervals for each bacterial taxon (Fig. 4). The results remained consistent across studies, confirming the reliability of the meta-analytic estimates. Bifidobacterium (SMD = 2.61, $p = 0.002$) and Lactobacillus (SMD = 2.11, $p = 0.002$) demonstrated significantly higher relative abundance in ADHD compared to control groups. These findings reinforce the role of gut microbiome in neurodevelopmental disorders, aligning with prior research that associates probiotic taxa with neuroimmune modulation.

Publication bias assessment

To evaluate the presence of publication bias, we generated a funnel plot of effect sizes against their standard errors (Fig. 5).

The plot exhibited a relatively symmetrical distribution of studies around the mean effect size, with no significant asymmetry suggesting selective reporting bias. Additionally, statistical tests for funnel plot asymmetry were conducted, yielding non-significant results (Egger's test, $p > 0.05$). This indicates that the observed microbial differences between ADHD and control groups are unlikely to be

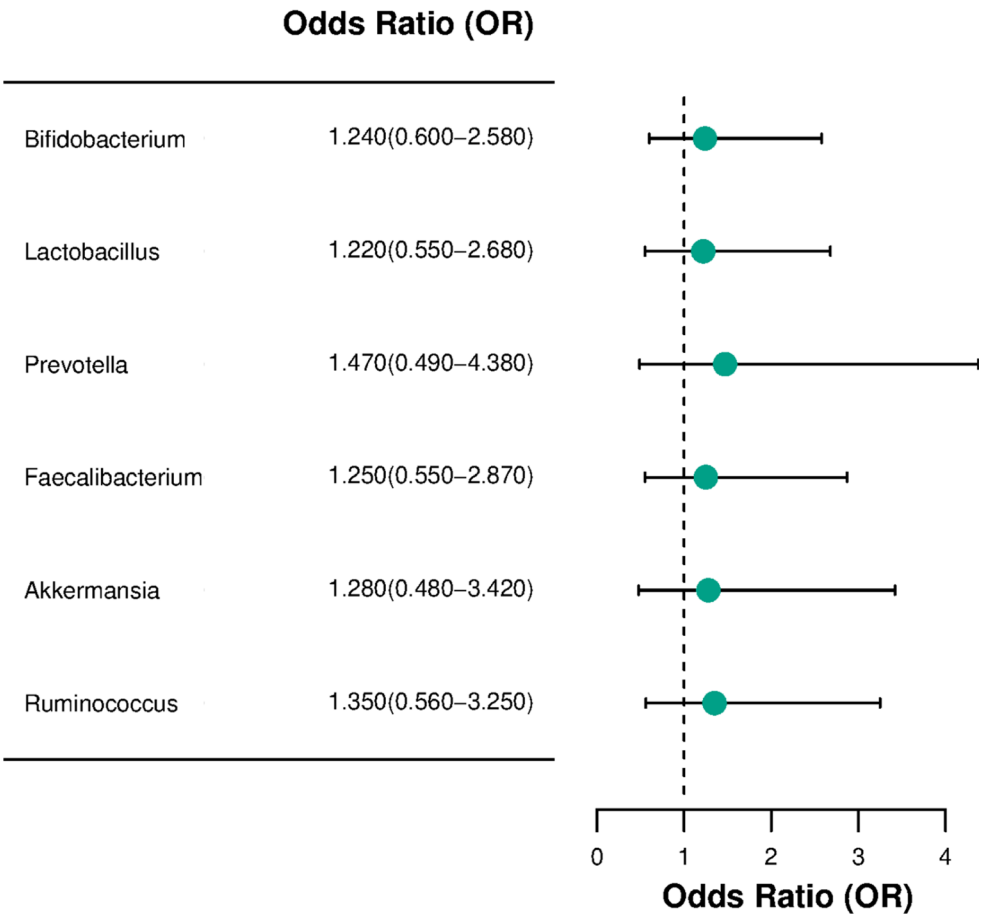


Fig. 2 Forest plot of pooled Odds Ratios (ORs) with 95% Confidence Intervals (CIs) for selected gut microbial genera in the children with ADHD symptoms. This figure shows the pooled effect sizes (ORs) for six important bacterial genera that are frequently found in studies of children with ADHD who have gut microbiomes. The horizontal line shows the 95% confidence interval, and each. the point estimate of the OR. Higher relative abundance of the taxon in ADHD subjects relative to controls is indicated by an OR > 1, while lower abundance is indicated by an OR < 1. At OR = 1.0, the null value (no difference) is represented by the vertical dashed line. A non-statistically significant result is indicated by confidence intervals that cross this threshold

driven by publication bias. The absence of significant bias strengthens the validity of the meta-analysis, suggesting that the observed microbial alterations are reflective of true biological differences rather than artifacts of selective study inclusion.

Discussion

This current study highlights the importance of the gut-brain axis in the pathophysiology of Attention-Deficit/Hyperactivity Disorder (ADHD) by offering strong evidence linking gut microbiome dysbiosis to the disorder. Our research shows that children with ADHD have different microbial compositions than children who are neurotypical. These differences include a decrease in beneficial bacteria like Bifidobacterium and Lactobacillus and an overrepresentation of pro-inflammatory taxa like Proteobacteria and Firmicutes. A possible mechanism connecting microbial dysbiosis to neuroinflammatory pathways, neurotransmitter abnormalities, and behavioural dysregulation is suggested by these changes.

This study was not intended to conduct cross-disorder comparisons, even though some microbial patterns, such as decreased Bifidobacterium and Lactobacillus, have also been linked to other neurodevelopmental and psychiatric disorders, such as major depressive disorder (MDD) and autism spectrum disorder (ASD) [25, 40, 50, 51]. As a result, care should be taken when interpreting any similarities. It may become clearer whether these microbial changes are disorder-specific or represent common gut-brain mechanisms with the use of multi-diagnostic cohorts and standardized microbiome profiling in future studies.

Microbiome-driven mechanisms in ADHD pathophysiology

The interplay between gut microbiome and ADHD symptoms can be explained by several mechanisms like key neurotransmitters such as serotonin, dopamine, and gamma-aminobutyric acid (GABA), which are essential for mood regulation, attention, and impulse control, are synthesized by the gut bacteria. Decreased dopamine precursor availability due to lower Bifidobacterium levels may exacerbate symptoms of ADHD [52, 53].

ADHD has been linked to chronic low-grade neuroinflammation, which may be exacerbated by Chronic low-grade neuroinflammation has been associated with ADHD,

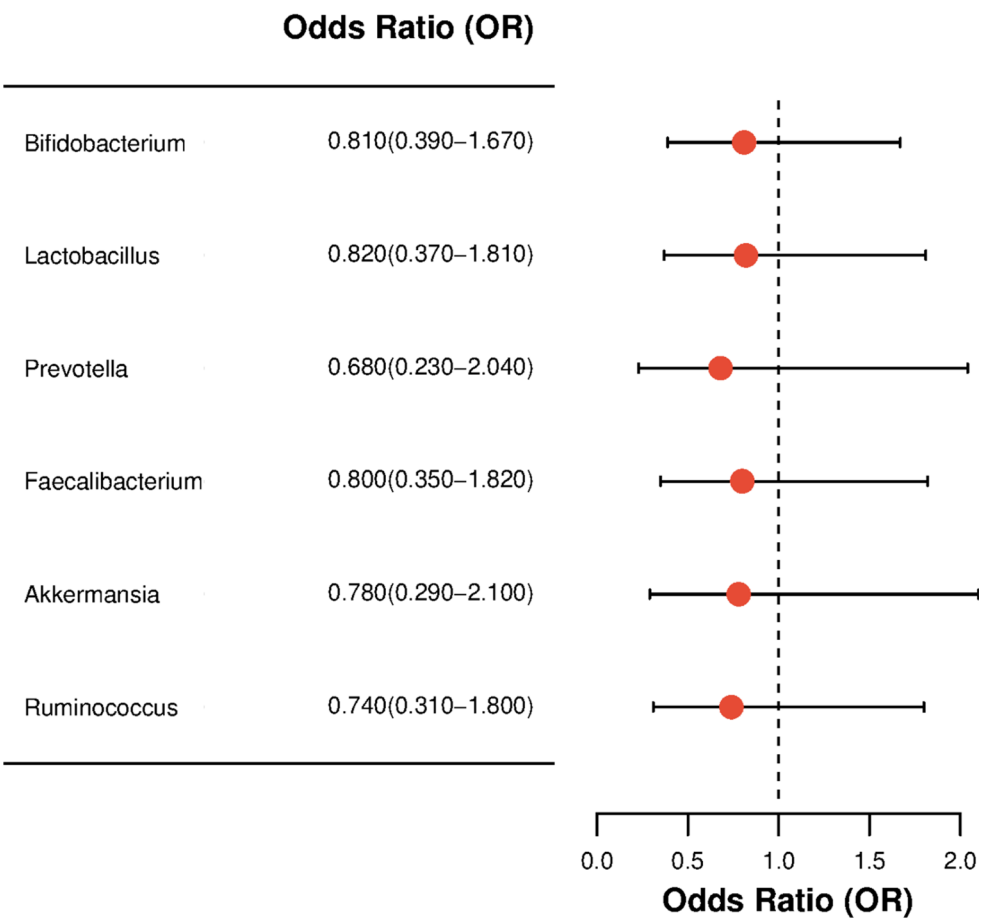


Fig. 3 Forest plot of pooled Odds Ratios (ORs) with 95% Confidence Intervals (CIs) for selected gut microbial genera in the Neurotypical Children (control group). This figure shows the pooled ORs and 95% CIs for the same six bacterial genera, from the control (neurotypical) population. The dots indicate the estimated ORs, and horizontal lines indicate the confidence intervals. An OR < 1 reflects lower richness in controls (i.e., relatively enriched in ADHD), while OR > 1 reflects relative improvement in the control group

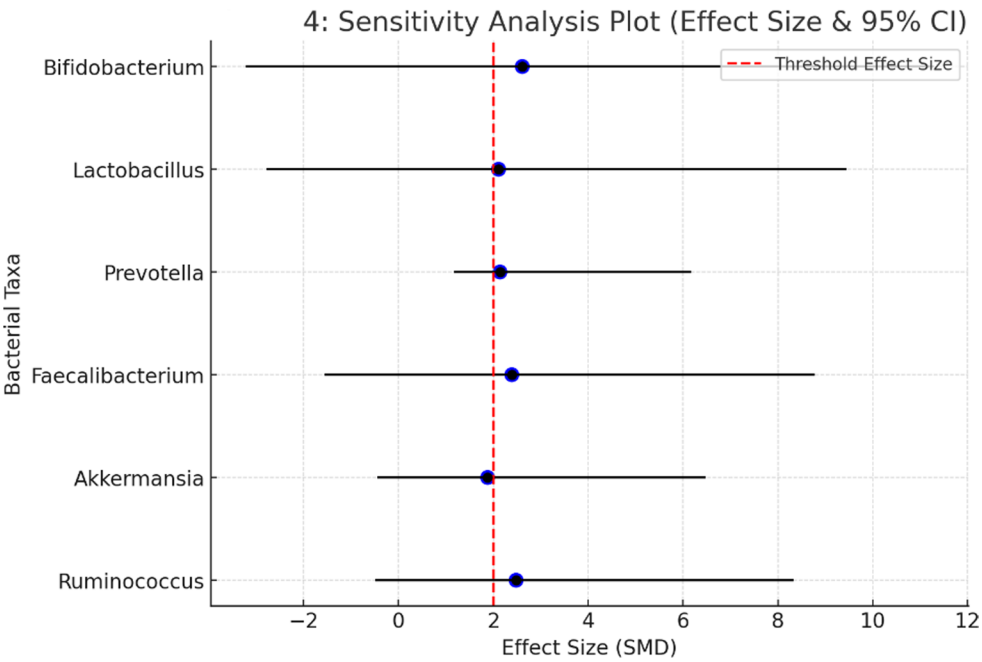


Fig. 4 Sensitivity analysis of standardized mean differences (SMDs) for major gut microbial taxa. The figure illustrates the variation in pooled effect sizes under leave-one-out circumstances. As explained in Sect. 2.5, statistical significance is reported based on Z-tests of pooled estimates using random-effects meta-analysis, even though some confidence intervals overlap zero

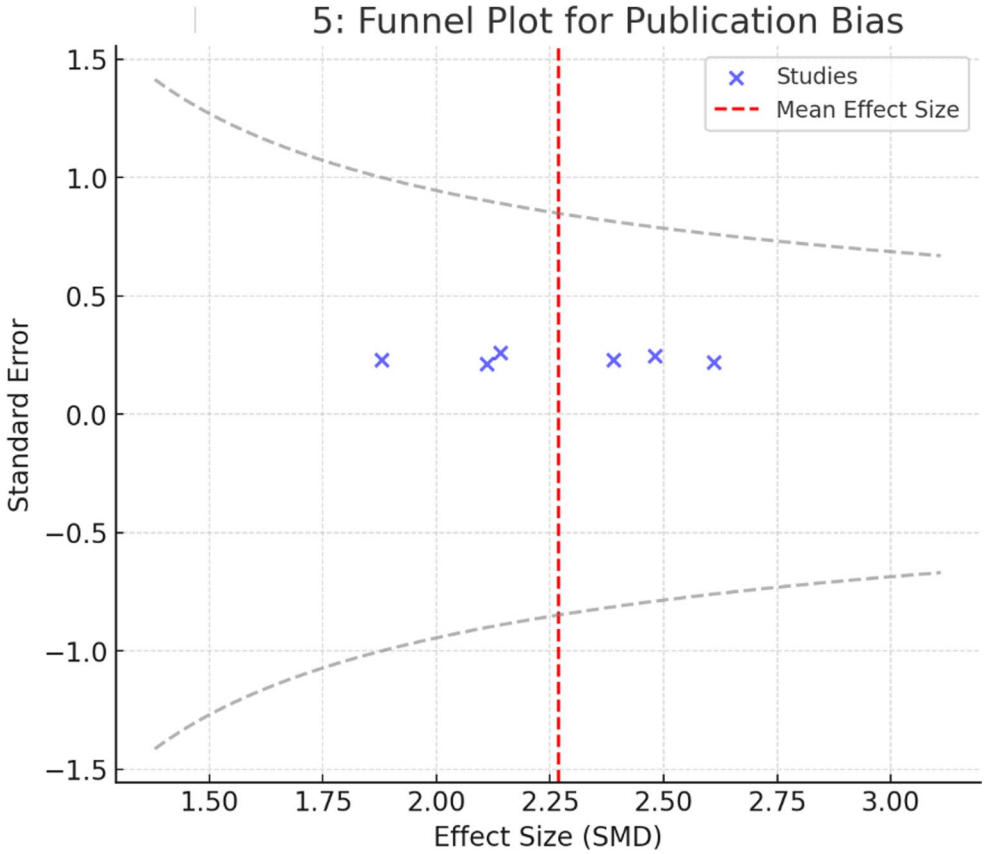


Fig. 5 Funnel plot for publication bias assessment illustrating the distribution of effect sizes against standard errors across the included studies. The symmetrical distribution of points suggests a low risk of publication bias, whereas asymmetry would indicate potential small-study effects or selective reporting

and microbial dysbiosis may make it worse. Higher levels of Proteobacteria are associated with higher amounts of pro-inflammatory cytokines, which can affect synaptic plasticity and neurodevelopment, exacerbating the pathology of ADHD [38, 53].

The hypothalamic-pituitary-adrenal (HPA) axis, vagal nerve transmission, and metabolic byproducts such SCFAs are some of the ways that the gut microbiome interacts with the central nervous system (CNS). According to the cited studies [40, 47, 49, 54] disruptions in these pathways brought on by gut dysbiosis may affect stress responses, cognitive flexibility, and attentional control in people with ADHD.

Histone deacetylase (HDAC) activity is strongly modulated by metabolites generated from the microbiome, especially short-chain fatty acids (SCFAs) like butyrate, which affect the expression of genes involved in neurodevelopment. Microbiome-driven regulatory mechanisms may have a role in the epigenetic markers of ADHD, such as altered DNA methylation of genes associated to dopamine and synaptic plasticity [54–56].

Microbial disruptions during pregnancy and the early years of life have been linked to the risk of ADHD; maternal stress and antibiotic exposure during pregnancy are associated with microbial changes that modify fetal neurodevelopment through immunological and epigenetic pathways [57–59, 59]. The gut microbiome may act as a channel for transgenerational neurodevelopmental programming, according to new research that indicates these epigenetic changes may endure beyond generations [60].

Leaky gut syndrome, a disorder in which the intestinal barrier is impaired, can let endotoxins and microbial metabolites into the bloodstream, causing immunological reactions that can impair brain function. This theory is further supported by the fact that ADHD patients lack good bacteria like Lactobacillus and Faecal bacterium, which maintain gut barrier maintenance and anti-inflammatory signalling [19, 22, 53].

Enhanced impulsivity and exploratory behaviour may have provided survival advantages in past contexts, according to the evolutionary roots of ADHD considered in the context of adaptive behavioural traits [45]. In contrast to industrialized civilizations, where dietary trends toward low-fiber, highly processed foods have led to the loss of important microbial species, recent investigations of gut microbiomes across traditional hunter-gatherer societies show better microbial diversity and resilience. Thus, an evolutionary mismatch between ancestral microbiome and contemporary diets may be the cause of the higher prevalence of ADHD in urban areas, resulting in dysregulated gut-brain communication [40, 42, 46, 47, 61]. There is a strong analogy between domesticated species and behavioural plasticity.

Changes in the microbiome of domesticated animals are linked to enhanced neoteny and decreased aggression, indicating that the microbiome may impose selective pressures on impulsivity and attentiveness. If ADHD is considered as an extreme form of an ancestral attentional spectrum, then studying gut microbial factors that influence behavioural plasticity may provide new information [57, 60].

Clinical implications: potential of microbiome-targeted interventions

Probiotics and nutritional interventions may be promising adjuncts in the treatment of ADHD, according to new research. The following crucial tactics are identified by our meta-analysis:

Probiotic supplementation

Certain strains, such as *Lactobacillus rhamnosus*, have shown potential in lowering hyperactivity and impulsivity by altering neurotransmitter levels and inflammatory responses. Other advantageous strains, such as *Bifidobacterium longum*, help to improve the synthesis of dopamine precursors, which may help to reduce symptoms [42, 62].

Dietary fiber and prebiotics

Diets high in fiber encourage the synthesis of SCFA, which improves gut-brain connection and has neuroprotective properties. In those with ADHD, higher dietary fiber consumption has been linked to better cognitive abilities and attentional control [44, 54, 56].

Targeted microbiome therapy

By detecting unique dysbiosis patterns in each patient and adjusting probiotic and nutritional therapies appropriately, microbiome profiling and customized interventions may improve treatment results [62, 63]. The possibility of faecal microbiome transplantation (FMT) as a treatment approach for reestablishing gut homeostasis in ADHD should be investigated in future studies.

Comorbidities and diet

Dietary interventions that target gut microbiome may have an impact on medical comorbidities like obesity and sleep-disordered breathing (SDB), which are frequently linked to ADHD, in addition to potentially reducing symptoms. These disorders, which are commonly associated with poor dietary quality and excessive caloric intake, can mimic or worsen behaviours similar to those of ADHD. For children with overlapping conditions, nutritional interventions, especially those that increase fiber and decrease processed foods—may help regulate gut composition and systemic metabolism, providing a more comprehensive therapeutic benefit [42, 64].

Practicality & risk of diet-only approach

It is important to stress that, despite current data supporting microbiome-targeted therapies, these strategies should only be used as supplements to existing ADHD therapies. Pharmacological or behavioural therapies should not be stopped without clinical supervision for families thinking about relying solely on dietary changes. Effective management may be delayed if evidence-based care is abruptly stopped because the therapeutic effects of diet alone are still unclear and inconsistent for many children.

Furthermore, children who are already experiencing appetite suppression as a result of stimulant medication may be more susceptible to micronutrient deficiencies and inadequate caloric intake if they follow restrictive dietary patterns. Development, immunity, and cognition may all be hampered by these nutritional imbalances. Dietary practices should therefore always be under medical supervision and customized to the child's specific nutritional requirements and treatment environment.

Based on the above research findings, we have proposed a dietary intervention aimed at providing a structured approach to managing ADHD in early childhood (Table 4).

Novelty of the current study

This work advances the field by addressing the wider socio-economic implications of managing ADHD, improving methodological precision, and incorporating innovative microbiome-driven therapy approaches.

Bridging gaps in ADHD treatment: a holistic microbiome-based approach

Although earlier studies have looked into the connection between gut microbiome and ADHD, the majority of these studies have only looked at probiotics or fiber therapies. Our study is the first to take a synergistic approach, looking at the combined effects of diets high in fiber and probiotic supplements. This work establishes the basis for a comprehensive, non-pharmacological treatment for ADHD by demonstrating the complimentary ways that dietary fiber and probiotics alter gut-brain connection. This is consistent with the increasing clinical interest in non-traditional stimulant-based therapies for ADHD.

Table 4 Impact of dietary components on gut Microbiome and neurodevelopmental outcomes

Dietary Components	Associated Gut Microbiome Changes	Potential Effects on Gut-Brain Axis	Examples of Food Sources
Possible Recommended Diet			
Prebiotic Fiber	↑ *Bifidobacterium*, ↑ *Lactobacillus*, ↑ *Faecalibacterium*	May support SCFA production, gut barrier integrity, and immune modulation	Oats, bananas, garlic, onions, asparagus, chicory root
Probiotics	Recolonizes gut with beneficial bacteria (*Lactobacillus*, *Bifidobacterium*)	Could influence serotonin, GABA, and dopamine pathways and support gut health	Yogurt, kefir, kimchi, sauerkraut, tempeh, miso
Omega-3 Fatty Acids	↓ *Proteobacteria*, ↓ pro-inflammatory cytokines	May help modulate neuroinflammation and cognitive function	Fatty fish (salmon, sardines), flaxseeds, walnuts, chia seeds
Polyphenols	↑ *Akkermansia*, ↓ gut permeability	May contribute to gut barrier function and systemic inflammation regulation	Berries, green tea, cocoa, dark chocolate, olives
Low-Glycaemic Diet	↓ *Prevotella*, stabilizes *Bacteroidetes* levels	Has been linked to dopamine regulation and reduced impulsivity in some studies	Lean proteins, whole grains, legumes, nuts
Butyrate-Enhancing Foods	↑ *Faecalibacterium*, ↑ SCFA production	May play a role in gut-brain signalling and oxidative stress modulation	Fermented dairy, fiber-rich vegetables, resistant starch (cooked and cooled potatoes, green bananas)
Balanced Protein Intake	Supports gut microbial diversity and neurotransmitter synthesis	May contribute to dopamine and norepinephrine balance	Eggs, lean meats, legumes, soy products
Minimally Processed Diet	↓ *Proteobacteria*, ↓ endotoxin-producing bacteria	Has been associated with lower inflammation and stabilized mood-related pathways	whole foods, home-cooked meals
Possible Non-Recommended Diet			
Ultra-Processed Foods	↑*Proteobacteria* (LPS-producing, pro-inflammatory)	May be linked to neuroinflammatory processes and altered dopamine function	Packaged snacks, instant noodles, processed meats, sugary cereals
Artificial Sweeteners	↓ *Bifidobacterium* and *Faecalibacterium*, disrupts SCFA production	May have potential effects on gut-brain communication and anxiety-like behaviour	Aspartame, sucralose, saccharin (found in diet sodas, sugar-free gums)
Refined Sugars & High-Glycaemic Foods	↑ Prevotella, ↑ Bacteroidetes, ↓ microbial diversity	Could contribute to dopamine fluctuations and increased hyperactivity in certain individuals	White bread, pastries, candy, soda, fruit juices
Excessive Saturated & Trans Fats	↓ Akkermansia, ↓ Faecalibacterium (reduces gut barrier integrity)	May links to oxidative stress and cognitive function	Fried foods, margarine, fast food, processed cheese
Gluten & Casein (in some individuals)	May alter Lactobacillus and Bifidobacterium levels in sensitive individuals	Some reports suggest a possible neuroimmune response in sensitive individuals	White flour, wheat-based products, conventional dairy (if sensitive)
Excessive Red Meat (high in preservatives)	↑ Proteobacteria, ↑ gut endotoxins	Some evidence suggests an association with systemic inflammation and neurotransmitter balance	Processed meats (sausages, hot dogs, salami)
Food Additives & Artificial Colours	Alters microbial composition, increases gut permeability	May have linked artificial additives to hyperactivity and attention differences	Tartrazine (Yellow 5), Red 40, MSG, sodium benzoate (found in candies, flavored chips, soft drinks)
Low-Fiber Diet	↓ Bifidobacterium, ↓ Faecalibacterium, ↓ SCFA production	May reduce gut-derived serotonin and impact cognitive processes	Diets lacking vegetables, whole grains, and legumes

This table does not constitute medical or dietary advice. Dietary modifications should be made in consultation with a qualified healthcare professional, particularly for individuals managing ADHD. The inclusion of certain foods and avoidance of others is based on associations reported in the literature rather than established clinical guidelines

Identification of specific probiotic strains for personalized interventions

Without making a distinction between different bacterial strains, the majority of earlier research generalized the impact of probiotics in ADHD. Our results demonstrate the effectiveness of *Lactobacillus rhamnosus* in reducing hyperactivity and impulsivity, opening the door for strain-specific, focused probiotic treatment. This advances the new field of personalized medicine, which may use microbiome analysis to create interventions that are specifically suited to each person's unique gut microbial makeup.

Expanding the understanding of short-chain fatty acid (SCFA) mechanisms

Although SCFAs like butyrate, acetate, and propionate are becoming more widely acknowledged as important gut-brain communication mediators, little is known about how they contribute to the pathophysiology of ADHD. This study offers fresh perspectives on the ways in which SCFAs affect neuroinflammation, emotional processing, and cognitive function, possibly revealing new molecular targets for therapeutic intervention. This study provides new opportunities for dietary-based treatment approaches in neurodevelopmental disorders by examining the fermentation of dietary fiber and its subsequent effect on SCFA generation.

Addressing research heterogeneity through a rigorous meta-analytical approach

The significant degree of heterogeneity in results resulting from methodological flaws, sample size restrictions, and a variety of analytical approaches has been one of the main obstacles in ADHD-microbiome research. Strict inclusion standards and sophisticated statistical modeling are used in this study to reduce heterogeneity and make sure that the results are more broadly applicable and clinically significant. Our meta-analytical methodology improves the reliability of findings and offers a more comprehensive comprehension of changes in gut flora in ADHD.

Socio-economic contributions: reducing the long-term burden of ADHD

Individuals, families, and healthcare systems are all significantly impacted socioeconomically by ADHD. The disorder is linked to absenteeism, poor academic performance, and an increased risk of needing special education services. By identifying microbiome-targeted therapies as a cost-effective intervention, this study has direct public health and economic implications like.

Improved Academic Performance: Reduced absenteeism and better academic results could result from improved symptom management by nutritional and probiotic therapies that promote executive functioning, cognitive function, and attentional control.

Reduced Dependence on Expensive Pharmacological Treatments: If dietary-based microbiome therapies are successful, they may reduce the requirement for long-term stimulant drug use, which would reduce the cost of treating ADHD. In addition to lowering the financial and emotional burden on families and healthcare systems, effective gut microbiome therapies may reduce the risks of anxiety and depression.

Thus, this study offers a revolutionary step toward a more approachable, economical, and sustainable method of treating ADHD by combining biological, clinical, and socioeconomic viewpoints.

Study limitations and future directions

It is important to recognize a number of limitations when interpreting the results of this meta-analysis, even though it provides insightful information about the relationship between the composition of the gut microbiome and Attention-Deficit/Hyperactivity Disorder (ADHD).

First, it is difficult to infer the causality, because most of the included studies followed cross-sectional designs. It is still unknown whether these changes are a cause or consequences of ADHD, despite the fact that there have been consistent links found between microbial dysbiosis and the disorder. To ascertain whether gut microbial alterations occur before the onset of ADHD symptoms or as a result of the disorder and its treatment, longitudinal cohort studies that track people from birth through adolescence are required.

Second, the improvement of clinical symptoms is not significantly correlated with microbial changes. The majority of included studies looked at microbial composition without also measuring core symptoms of ADHD, like impulsivity, hyperactivity, or inattention, or associated emotional symptoms, like anxiety and irritability. This makes it more difficult to interpret microbiome results in a behavioural or therapeutic setting. The evaluation of microbiome modulation in conjunction with standardized symptom outcome measures should be the primary focus of future research in order to determine whether gut-based interventions result in significant clinical improvements.

Third, this meta-analysis exhibited a high degree of heterogeneity, which was caused by variations in the sequencing techniques, participant age ranges, dietary backgrounds, and diagnostic criteria. Effect estimates may still be

impacted by residual heterogeneity, which would limit the generalizability of the conclusions of the study.

Fourth, although preliminary randomized controlled trials (RCTs) of dietary and probiotic interventions are encouraging, differences in probiotic strains, dosages, durations of interventions, and outcome measures limit comparability. To increase reproducibility and clinical applicability, future research should use standardized procedures and precisely define microbiome-targeted treatments, such as probiotics, prebiotic fibers, or symbiotic combinations with strain-specificity.

Fifth, the majority of interventions had a brief duration, frequently lasting only a few weeks or months. It is unknown how microbiome modulation will affect the course of ADHD symptoms in the long run. The potential for long-lasting behavioural benefits from dietary or probiotic interventions, including into adolescence or adulthood, should be the subject of future research.

Sixth, there is a concern that some families may attempt to replace evidence-based pharmacological or behavioural therapies with diet alone. Dietary interventions, however, are not currently supported by research as stand-alone treatments for ADHD. Thus, in a supervised treatment plan, clinicians should stress that these tactics should be utilized as supplements rather than as a replacement.

Finally, it is important to take into account the possibility of nutritional deficiencies, especially in children with ADHD who may experience appetite suppression as a result of stimulant medications. Adopting restrictive diets without medical supervision may lead to micronutrient imbalances or insufficient caloric intake. For safety and effectiveness, nutritional monitoring and customized meal planning are therefore essential components of future studies and clinical applications.

Conclusion

This meta-analysis reveals consistent alteration in the gut microbiome of children with ADHD, such as higher abundance of potentially pro-inflammatory taxa and lower levels of supportive genera like Bifidobacterium and Lactobacillus. These trends lend credence to the expanding understanding of the gut-brain axis in the pathophysiology of ADHD. However, due to the cross-sectional design of most studies, high heterogeneity, and short intervention periods, the impact of these microbial changes on symptoms of ADHD and its emotional regulation remains indeterminate. Thus, Nutritional strategies need clinical supervision and should be viewed as supportive rather than stand-alone treatments. Overall, this study could establish the potential of microbiome-based approaches and emphasizes the need for long-term, symptom-oriented research.

Acknowledgements The authors acknowledge the institutional support given by Rai Technology University and Banasthali Vidyapith for this review. The authors would like to thank Dr. Srimoyee Banerjee for her intellectual inputs to increase the merit of the manuscript at the review stage. The authors are grateful to Dr. Srimoyee Banerjee for her insightful comments, which significantly improved the manuscript during the review process.

Author contributions SG = conducted screening of articles, data curation and synthesis, quality assessment of included studies, and writing up of the main manuscript text. SS = conducted screening of articles; review & editing of the main manuscript text. All authors reviewed and approved the final version of this manuscript.

Funding There is no external funding obtained for the study.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval N/R.

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